
Separating Drift and Respiratory Phase in Sparse Gaussian Processes for Chemical Sensor Reconstruction

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Abstract

Periodic sampling superposes an apparatus rhythm on the evolving response of a chemical sensor. We construct a compact Gaussian-process covariance that separates drift from signed respiratory-phase dependence and allows the oscillation amplitude to change during recovery. Exact block-sparse inference uses 500,000 measured observations from 928 sensor trajectories in 58 exposure experiments. On 9,280 frozen targets within excluded 12-second intervals, the adapted covariance achieves RMSE 0.00310 and CRPS 0.00276, reducing these scores by 69.3% and 65.5% relative to a fitted single-scale Wendland covariance. Removing phase dependence degrades both scores; a strong common-kernel Vecchia baseline remains competitive. The learned 13.584-second rhythm is corroborated by training-only spectral diagnostics, including air controls, and becomes more coherent during recovery. This supports waveform reconstruction but cautions against interpreting periodicity as evidence of chemical presence. Full calibration curves expose substantial class-dependent uncertainty errors despite aggregate 95% coverage of 96.86%. Exact conditioning preserves the specified model; it does not validate its noise assumptions or replace convergence diagnostics.

1 Introduction

An actively sampled chemical sensor measures both its environment and the dynamics of the sampling apparatus. Periodic ventilation produces a repeating resistance response, while adsorption, chamber transport, and the removal of the analyte source change its slowly varying envelope. Reconstructing an interruption that spans a nominal ventilation cycle therefore requires more than smooth interpolation in elapsed time. A model must distinguish local drift from phase-dependent recurrence and must express uncertainty where those components are weakly identified.

We study this problem using measured metal-oxide sensor responses to acetone and ethanol under flow modulation [Ziyatdinov et al., 2015a,b]. Our target is the missing waveform within an observed exposure experiment, not the concentration of an unseen chemical mixture. We construct a compact covariance with separate drift and oscillatory supports, signed phase dependence, and an exposure-to-recovery amplitude gate. The kernel belongs to the broad closure-based design space of Gaussian processes; the contribution is its domain-specific specification, exact support-aware implementation, and controlled evaluation on missing respiratory cycles.

Compact covariance and distributed matrix assembly make large exact Gaussian processes possible [Furrer et al., 2006, Noack et al., 2025, Risser et al., 2025]. Exactness here means inference under the covariance actually specified, to floating-point accuracy. It neither guarantees that the covariance describes the experiment nor eliminates uncertainty about its fitted parameters. This distinction is particularly relevant for sensor records: many measurements can be collected from comparatively few independent exposures. We retain this distinction throughout, reporting the number of scalar observations separately from the number of experiments.

Spectral kernels already establish the value of oscillatory covariance for pattern extrapolation [Wilson and Adams, 2013]; nonstationary compact kernels provide a complementary route to sparse exact inference [Noack et al., 2025]. Our construction combines these ideas with the exposure protocol, without asserting a new universal family of positive-definite functions. We compare it with the published single-scale Wendland kernel, a phase-removed ablation, and an observed-first Vecchia approximation using the same adapted covariance [Katzfuss and Guinness, 2021]. A common held-out interval design and full predictive scoring separate covariance specification, inference approximation, and calibration.

2 Methodology

2.1 A compact covariance on exposure trajectories and respiratory phase

Let $x = (j, t)$ identify a sensor–experiment trajectory j and elapsed time t . Write $W(q) = (1 - q)_+^8(35q^3 + 25q^2 + 8q + 1)$ for the polynomial used by Noack et al. [2025]. The existing-kernel comparator is

$$k_W(x, x') = \mathbb{K}_{j=j'} v W(|t - s|/r). \quad (1)$$

We introduce an exposure gate $h(t) = \{1 + \exp[-(t - 180)/2]\}^{-1}$ and $a_\eta(t) = \exp\{\eta h(t)\}$. Our respiratory covariance is

$$k_R(x, x') = \mathbb{K}_{j=j'} \left[v_d W\left(\frac{|t - s|}{r_d}\right) + v_o a_\eta(t) a_\eta(s) W\left(\frac{|t - s|}{r_o}\right) \cos\left(\frac{2\pi(t - s)}{P}\right) \right]. \quad (2)$$

The variances v_d, v_o , supports r_d, r_o , period P , and observation-noise standard deviation are positive. The gate coefficient η controls the recovery-to-exposure oscillatory amplitude ratio. The transition time follows the acquisition protocol; its smoothing width is fixed, not inferred from test observations. We fit the remaining parameters in log coordinates, except η .

The cosine kernel is the inner product of $(\cos(2\pi t/P), \sin(2\pi t/P))$ and its counterpart at s . Multiplying it by a valid temporal taper and by $a_\eta(t)a_\eta(s)$ preserves positive semidefiniteness. Adding the drift term and taking a direct sum over trajectories also preserves this property. Unlike a single nonnegative distance taper, the oscillatory component can correlate opposite phases negatively while retaining positive same-phase dependence. This is a specific extension within the arbitrary-core design space of gp2Scale, not a proof that all of its possible user-defined kernels are inadequate. The phase-removed ablation replaces the cosine by one and otherwise retains the two scales and the gate.

2.2 Exact assembly, likelihood, and prediction

For standardized training responses z , the covariance is $C = K + \sigma^2 I$. We evaluate

$$\ell(\theta) = -\frac{1}{2} z^\top C^{-1} z - \frac{1}{2} \log |C| - \frac{N}{2} \log(2\pi). \quad (3)$$

Dask workers evaluate every pair within each trajectory and return sparse coordinate blocks; the host assembles the complete compressed sparse row matrix. Different trajectories have identically zero covariance by the model, so their blocks can be omitted algebraically before kernel evaluation. This support rule reduces candidate work to $\sum_j n_j^2$ without discarding any nonzero covariance. It differs from assuming that numerically small entries can be deleted after a dense-kernel computation.

The resulting global covariance is a direct sum. We factor each complete component by Cholesky and concatenate its solves, with $\log |C| = \sum_j \log |C_j|$. Dense arithmetic within a complete component is an exact implementation of this block-sparse factorization. All scalar observations contribute to every full-data objective. Thus the computational scale is a large joint likelihood with shared parameters, not a single densely coupled matrix of the same dimension. The experiment does not establish scalability to such a densely coupled problem.

We initialize covariance parameters on a training-only pilot, refine them with the full-data likelihood, and use the supplied fvGP block Metropolis–Hastings implementation. Analytic gradients for initialization use

$$\partial_q \ell = \frac{1}{2} \text{tr} [(\alpha \alpha^\top - C^{-1}) \partial_q C], \quad \alpha = C^{-1} z. \quad (4)$$

The blocks separate drift parameters, oscillation and gating, and noise. Predictions condition on the final accepted parameter state. For test inputs X_* we use the exact component solves in

$$\mu_* = K_{*X}\alpha, \quad V_* = K_{**} + \sigma^2 I - K_{*X}C^{-1}K_{X*}. \quad (5)$$

The reported marginal variances include observation noise. They do not integrate hyperparameter uncertainty, and short chains are used as likelihood refinements rather than evidence of converged posterior sampling.

2.3 A common-kernel approximation

The response-Vecchia baseline replaces the joint density by $\tilde{p}(z) = \prod_i p(z_i | z_{g(i)})$, where $g(i)$ contains the preceding observations within the same chronological trajectory. Each conditional uses the exact covariance in Eq. (2). At prediction time, test nodes follow all training nodes and condition on the nearest observed training parents on either side of the gap. This is an approximation to the conditioning set, not a different kernel. Its optimizer uses only training responses; validation selects prediction-parent configurations. Test responses enter only the final scoring.

3 Experimental Setup

3.1 Chemical measurements and excluded intervals

We retrieved the UCI Gas Sensor Array under Flow Modulation archive, retaining the released normalized resistance change $(R(t) - R_0)/R_0$ [Ziyatdinov and Fonollosa, 2015]. It comprises 58 exposure experiments, 16 sensors, and 928 trajectories. The raw array contains 6,960,000 scalar measurements. We map sample indices to elapsed time using the documented 25 Hz acquisition rate over 300 seconds. We treat the original acquisition and baseline normalization as provided preprocessing; we apply no label smoothing or synthetic augmentation.

Using seed 20261004, we first select a contiguous 12-second test interval in every trajectory and sample 10 target times inside each interval. Every observation within that interval, including unscored times, is excluded from fitting. We next select disjoint validation intervals by the same procedure and only then sample 500,000 training measurements without replacement from the remaining observations. The frozen test and validation sets each contain 9,280 observations. Training and test identities are saved, hashed, and disjoint. Per-trajectory response means and standard deviations are computed using training observations alone; predictions are transformed back to the released resistance units before scoring. Validation is used only for configuration diagnostics, never as training data.

The missing-interval design tests reconstruction across an interruption rather than interpolation between almost adjacent measurements. It does not evaluate transfer to a new exposure, a new sensor, a new chemical, or a future acquisition day. Because sensors within an exposure share experimental conditions, uncertainty on differences in predictive scores uses a paired bootstrap over complete experiments, retaining all sensor trajectories in each sampled experiment.

3.2 Model fitting and scoring

The exact comparisons use the same trajectory factorization, data, preprocessing, Gaussian observation model, and test points. Support bounds and all seeds were recorded before the full-scale runs. The full-data trace includes every covariance objective, solve residual, log determinant, parameter state, and accepted MCMC block. Training workers have one numerical-library thread each; the number of workers and sampled peak resident memory are reported with the results. Vecchia is repeated with seeds 11, 22, and 33, which perturb a shared training-derived initializer by Gaussian noise of standard deviation 0.015 in parameter coordinates. The preregistered comparison uses the same 500,000-point pool. A disclosed extension also fits all 6,403,200 measurements outside the original test and validation gaps, the largest training set available in the archive under this exclusion rule. This changes the amount of training data while preserving the test task; it is reported separately.

We report RMSE and the Gaussian continuous ranked probability score (CRPS) [Gneiting and Raftery, 2007]. With $u = (y - \mu)/s$, the latter is

$$\text{CRPS} = s \left[u \{ 2\Phi(u) - 1 \} + 2\phi(u) - \pi^{-1/2} \right]. \quad (6)$$

Table 1: Reconstruction on the same 9,280 frozen targets. Coverage is empirical 95% coverage. Vecchia rows give means over three seeds; complete seed scores and sample standard deviations are reported in the text and appendix. Time covers the final fitting/refinement and prediction run. Memory is the largest sampled run peak; shared initialization costs are additional.

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak memory (GB)
gp2Scale	Respiratory	500,000	9,280	0.00310	0.00276	96.86%	0.230	95.37
gp2Scale	Wendland Phase	500,000	9,280	0.01009	0.00798	96.44%	0.182	44.36
gp2Scale	removed	500,000	9,280	0.00807	0.00617	96.54%	0.276	118.08
Vecchia	Respiratory	500,000	9,280	0.00352	0.00288	96.86%	0.174	0.55
Vecchia	Respiratory	6,403,200	9,280	0.00414	0.00289	94.15%	0.537	0.76

Coverage is the empirical fraction inside a central Gaussian prediction interval. The main table reports 95% coverage; the calibration figure reports the complete predefined grid of nominal levels. Lower RMSE and CRPS are better; coverage should track its nominal level. We also report exposure/recovery and gas-class breakdowns, including unfavorable comparisons.

3.3 Reference implementation and numerical checks

We reproduce the provided MNIST binary-regression construction for the indicator of digit five, independently splitting the combined archives into 60,000 training and 10,000 test images for three seeds. Normalized pixels use Manhattan distances and a Wendland times delta-membership covariance. Since the source paper does not specify its exact delta sets, training-only clustering supplies those memberships. The preregistered absolute tolerance on the mean Brier score is 0.006 around the published 0.018. This is a coarse score-level reproduction, not an identical-protocol replication; its covariance diagnostics are reported in Appendix A.

Separately, we compare our likelihood and predictive moments against the provided fvGP build with its gp2Scale dispatcher and sparse-LU backend, dense Cholesky, and Vecchia with every predecessor retained. The saved audit also compares analytic derivatives against central finite differences. These checks distinguish implementation errors from the modeling deficiencies of a particular benchmark kernel.

4 Experimental Results

4.1 Reconstruction across a nominal cycle

Table 1 compares the complete training-scale fits on identical held-out observations. In the matched-data comparison, the adapted covariance improves RMSE from 0.01009 to 0.00310 and CRPS from 0.00798 to 0.00276. Paired experiment-cluster bootstrap intervals for the adapted-minus-Wendland differences are $[-0.00878, -0.00514]$ for RMSE and $[-0.00616, -0.00428]$ for CRPS. Both exclude zero. These comparisons support the inadequacy of this fitted single-scale kernel for the present reconstruction task; they do not establish inadequacy of every arbitrary-core construction allowed by gp2Scale. Appendix C also checks clustering by acquisition batch.

The phase-removed model retains the two supports and the exposure gate but obtains RMSE 0.00807 and CRPS 0.00617. Thus the gain cannot be explained solely by adding a second scale and a nonstationary amplitude. Representative, deterministically selected reconstructions appear in Figure 1. Every selected example is the first trajectory in its broad gas class, rather than an example chosen for its error reduction.

The three-seed matched-data Vecchia results are RMSE 0.00352 ± 0.00028 and CRPS 0.00288 ± 0.00022 . The exact adapted model has lower RMSE than each of these runs, but the best Vecchia seed has slightly lower CRPS. We therefore find a much stronger advantage for the covariance adaptation than for exact conditioning over a well-configured approximation. Since each method also fits its own likelihood, remaining differences combine inference and hyperparameter-fitting effects. Appendix C includes every seed, initialization attempt, and subgroup score.

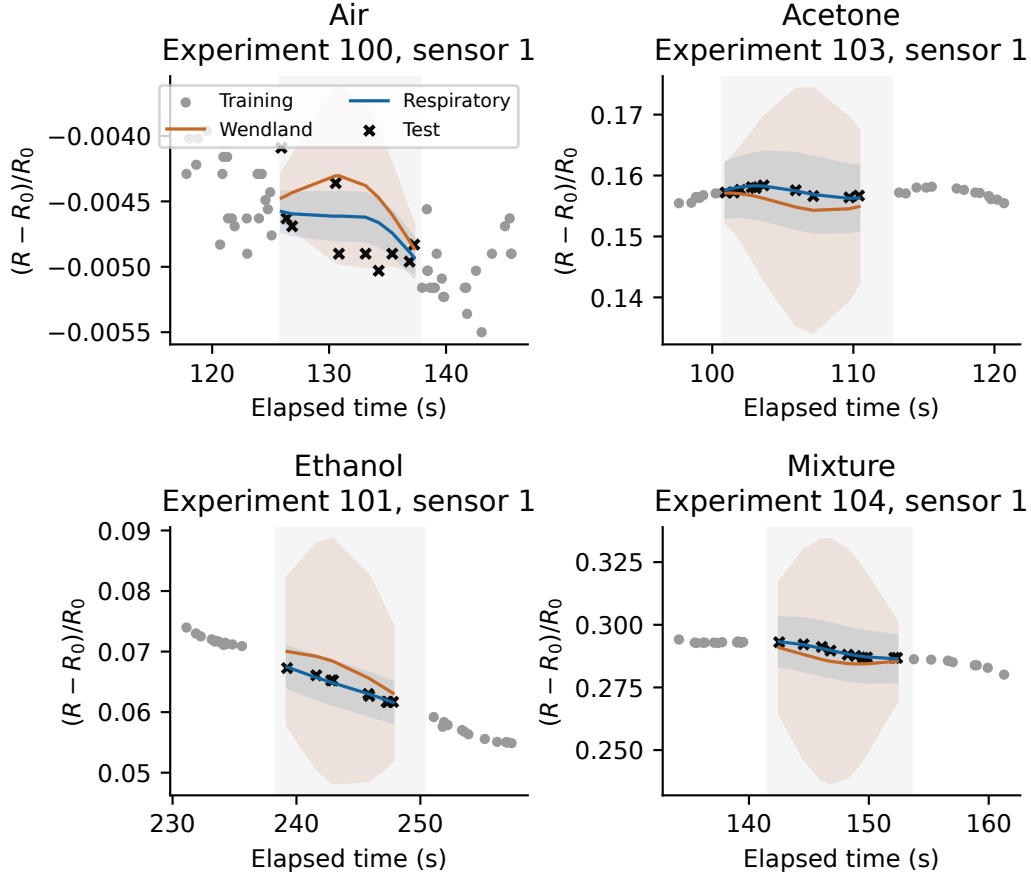


Figure 1: Held-out means and 95% predictive intervals in four predetermined trajectories. Shading marks the complete excluded interval; curves connect predictions at the scored test times. Training measurements outside the interval are gray, and held-out measurements are black. The examples include an air control.

Using all 6,403,200 eligible measurements, validation-selected Vecchia checkpoints obtain RMSE 0.00414 ± 0.00028 and CRPS 0.00289 ± 0.00006 , with mean 95% coverage of 94.15%. These fits use 128 chronological training parents and 512 observed prediction parents. Validation selected the noise-fitted checkpoint because subsequent joint tuning degraded its score. At the higher sampling density, the median span of 128 training parents shrinks from 71.00 to 5.12 seconds; sample count and conditioning context both change. Appendix D reports the full joint fits, checkpoint selection, evaluation counts, and stopping reasons.

4.2 Calibration and computational evidence

Aggregate 95% coverage is 96.86% for the adapted model and 96.44% for Wendland. This apparent agreement with the nominal level conceals a severe calibration defect: the adapted model’s nominal 50% interval contains 91.73% of observations. Figure 2 shows overcoverage at central levels across the compared models. Furthermore, its 95% coverage is only 78.98% in air controls but exceeds 99.6% in each chemically loaded gas class. A single aggregate coverage number would conceal this reversal.

The adapted full covariance contains 170,909,842 nonzero entries, or 0.0684% of the global matrix. Within the 269,842,238 candidate pairs belonging to the same trajectory, however, 63.34% remain nonzero. Most global sparsity therefore comes from the explicit trajectory-independence assumption. Figure 3 distinguishes the signed oscillatory component from the support of the complete covariance.

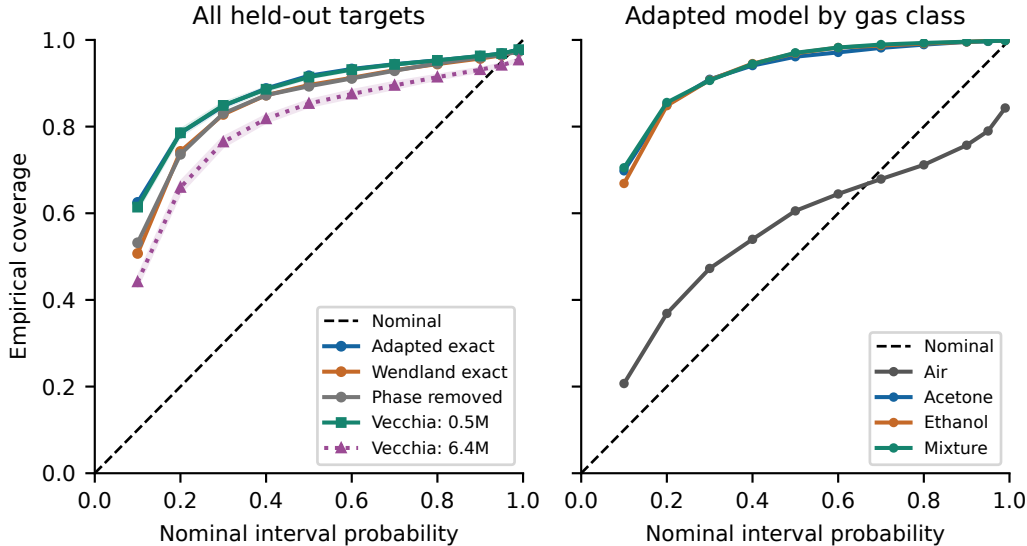


Figure 2: Left: probability-coverage curves on the same held-out observations, including both Vecchia training scales. Right: the adapted model stratified by gas class. The diagonal denotes calibration. Vecchia bands show sample standard deviations across three seeds, not confidence intervals over independent experiments.

Both fitted supports reach their prescribed upper bound; the support percentages are not evidence that a unique physical correlation horizon was identified.

All full-data likelihood evaluations include the solve and log determinant. The final standardized-data log marginal likelihood is 734,975.890, and the final adapted fitting/refinement and prediction run takes 13.79 minutes. Initializer construction is additional work and is disclosed in Appendix E; the final-run timing is not a cold-start benchmark. Chain traces, per-block acceptance, and the full numerical audit are provided in Appendices A and B.

5 Analysis

5.1 Mechanism: phase information rather than sparsity alone

An excluded interval can separate observations that occupy similar respiratory phases. A single distance taper cannot distinguish that relationship from a separation of half a cycle. Equation (2) supplies this distinction without sacrificing a long drift component. The phase-removed ablation is worse on both primary scores, and its paired bootstrap intervals also exclude zero (`logs/paired_bootstrap.json`). This is empirical evidence for the phase term beyond an extra scale and the known exposure boundary.

The support perturbations provide a more direct control. Holding the oscillatory support at its fitted value while setting the drift support to 30, 60, or 120 seconds leaves exactly the same 170,909,842 nonzeros, yet RMSE changes from 0.00970 through 0.00396 to 0.00310. The weights and decomposition of the covariance, not its sparsity percentage alone, determine reconstruction quality. Increasing drift support to 240 seconds yields lower CRPS but worse RMSE and more nonzeros. These fixed-parameter comparisons are sensitivity diagnostics, not separately optimized alternatives; all their scores are reported in Appendix E.

5.2 A shared rhythm persists after removal of the analyte source

The fitted effective period is 13.584 seconds, with a recovery-to-exposure oscillation-amplitude ratio of 0.826 and variance ratio of 0.682. These parameters summarize statistical covariance in the released resistance traces. In particular, the effective period differs from the nominal 12-second

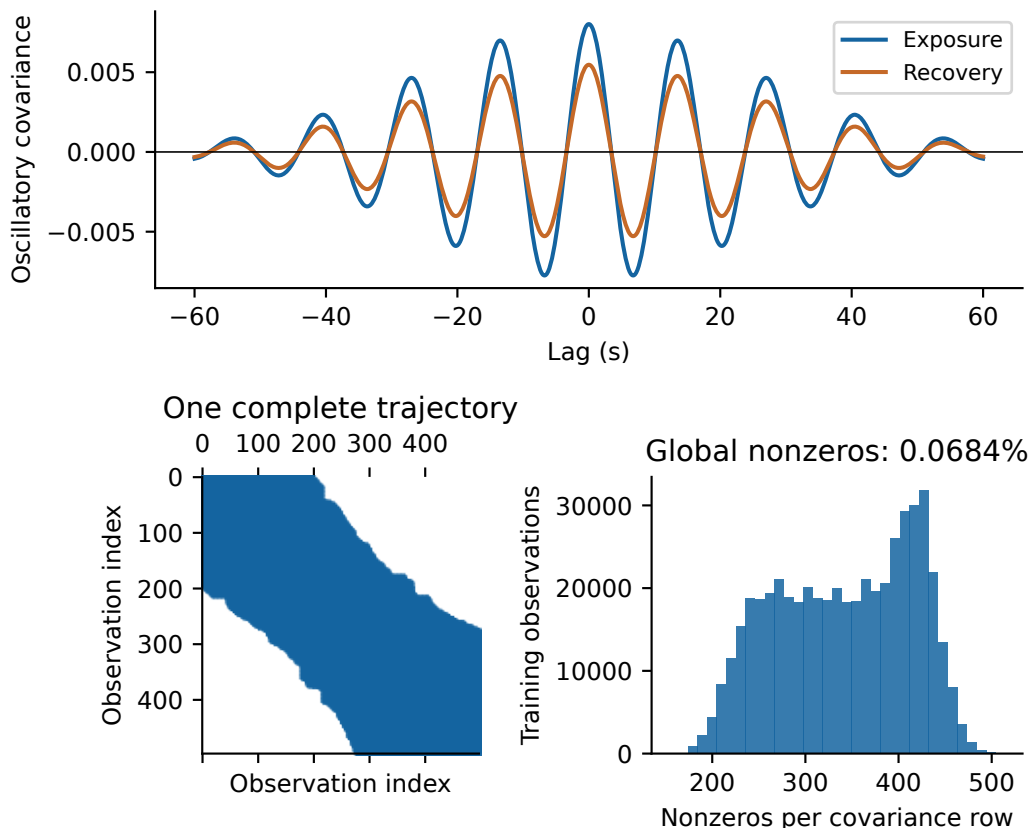


Figure 3: Top: the learned oscillatory covariance component, which changes sign with phase and has a smaller recovery amplitude. This is a component of the total covariance, not a claim that all displayed global nonzeros are negative. Bottom left: the actual nonzero pattern of one complete trajectory covariance. Bottom right: row degrees in the assembled full-scale covariance.

ventilation setting; a simultaneously measured flow-reference channel would be needed to attribute that difference.

Independent training-only harmonic regressions corroborate the rhythmic structure (Figure 4). Air controls have a peak near 13.55 seconds in both phases, explaining approximately 0.582 and 0.586 of their detrended variance. In acetone, ethanol, and mixtures, the peak explained fractions during exposure are 0.153, 0.269, and 0.240; during recovery they increase to 0.708, 0.716, and 0.716. For example, the experiment-cluster bootstrap interval for acetone recovery is [0.674, 0.741], compared with [0.141, 0.167] during exposure. The underlying spectra and bootstrap replicates are summarized in `logs/training_spectrum_summary.json` and `logs/spectral_bootstrap.json`.

Greater fractional coherence during recovery does not imply larger oscillation amplitude. Training-only amplitude estimates decrease in the chemically loaded trajectories, while the air-control amplitude is nearly unchanged (`logs/domain_analysis.json`). The combined evidence is consistent with a persistent sampling-apparatus component becoming more prominent relative to the changing chemical transient. It does not identify adsorption kinetics or establish causation. This distinction matters operationally: the periodic component is valuable for filling missing sensor readings, but its presence alone is insufficient for inferring a chemical exposure. The original studies establish the value of flow modulation for sensing [Ziyatdinov et al., 2015a,b]; the present evidence specifically characterizes its consequences for missing-cycle reconstruction and recovery-phase uncertainty.

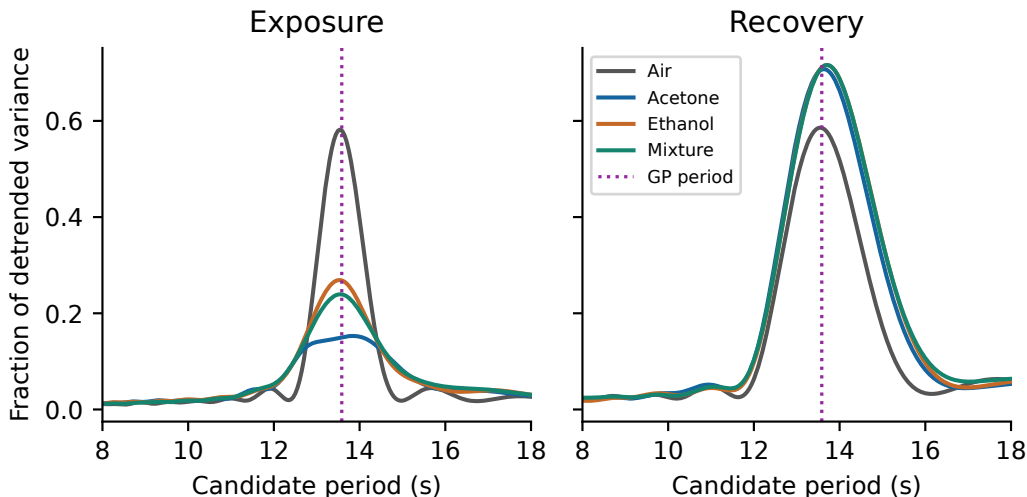


Figure 4: Independent training-only spectral diagnostics after cubic detrending. Curves average the explained harmonic variance fraction within each gas class. The vertical line is the fitted GP period. A pronounced rhythm in air and recovery demonstrates that periodic structure alone is not specific evidence of analyte presence.

5.3 Failure modes and limitations

Exact conditioning does not resolve the class-dependent coverage errors. Air controls are the strongest uncertainty failure even though their absolute RMSE is small; chemically loaded traces have overly conservative central intervals. The largest mean predictive standard deviation occurs in mixtures (0.01567), compared with 0.000587 in air, so absolute interval width alone does not identify the worst coverage. These regional moments are saved in `logs/subgroup_metrics.json`. The pooled standardized-noise model and remaining waveform misspecification cannot be distinguished as causes with this design. Moreover, trajectory-level CRPS gain does not increase monotonically with harmonic variance fraction: the observed Spearman correlation is -0.211 (Appendix F). A periodicity measure alone is therefore an inadequate rule for assigning reconstruction confidence.

Both support parameters are boundary estimates. The final chain accepts no proposals in either covariance block and only some noise proposals, so it does not establish posterior convergence or uncertainty in the learned period and supports. Our operative fitting procedure is bounded full-data maximum likelihood with logged Metropolis checks, and our predictions are conditional on the fitted state. The numerical checks establish exact computation for that model, not a fully Bayesian analysis. No explored support configuration failed for lack of memory; we do not infer an unmeasured memory frontier from these runs.

There are only 58 exposure experiments, collected in five batches, and correlation between sensors in one exposure is omitted from the prior. Experiment-cluster bootstrap intervals assume independence between exposures; the batch-level sensitivity checks relax that assumption but have very few clusters. Neither repairs a misspecified predictive covariance. The results concern within-record missing intervals, not forecasting or transfer to unseen mixtures. The larger Vecchia runs use all available nonexcluded observations, but their bounded optimization and fixed training parent count do not exhaust possible approximate inference configurations. Finally, the common-kernel baselines are favorably initialized with exact-model training checkpoints, and optimization remains sensitive to the starting mode. These qualifications limit claims about cold-start efficiency and intrinsic superiority of exact inference.

6 Conclusion

Separating drift from respiratory phase materially improves the reconstruction of missing chemical-sensor waveforms at a jointly fitted scale of 500,000 observations. The phase-removed control and

fixed-support perturbations show that covariance structure, rather than sparsity percentage alone, accounts for the improvement over the existing single-scale kernel. A strong approximate baseline retains much of that benefit.

The application also exposes a distinction between reconstructing a waveform and interpreting its origin. A coherent rhythm persists in air controls and becomes fractionally stronger during recovery, while uncertainty coverage remains sharply class dependent. The resulting model is useful for this interpolation task, but it does not identify chemical presence, validate a physical kinetic mechanism, or supply converged hyperparameter uncertainty. Exact Gaussian-process arithmetic makes these modeling questions accessible at scale; it does not answer them automatically.

References

- Reinhard Furrer, Marc G. Genton, and Douglas Nychka. Covariance tapering for interpolation of large spatial datasets. *Journal of Computational and Graphical Statistics*, 15(3):502–523, 2006. doi: 10.1198/106186006X132178.
- Tilman Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- Matthias Katzfuss and Joseph Guinness. A general framework for vecchia approximations of gaussian processes. *Statistical Science*, 36(1):124–141, 2021. doi: 10.1214/19-STS755.
- Marcus M. Noack, Mark D. Risser, Hengrui Luo, Vardaan Tekriwal, and Ronald J. Pandolfi. gp2scale: A class of compactly-supported non-stationary kernels and distributed computing for exact gaussian processes on 10 million data points. *arXiv preprint arXiv:2512.06143*, 2025. URL <https://arxiv.org/abs/2512.06143>.
- Mark D. Risser, Marcus M. Noack, Hengrui Luo, and Ronald Pandolfi. Compactly-supported nonstationary kernels for computing exact gaussian processes on big data. *Environmetrics*, 2025. doi: 10.1002/env.70054. URL <https://arxiv.org/abs/2411.05869>.
- Andrew Wilson and Ryan Adams. Gaussian process kernels for pattern discovery and extrapolation. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 1067–1075, 2013. URL <https://proceedings.mlr.press/v28/wilson13.html>.
- Andrey Ziyatdinov and Jordi Fonollosa. Gas sensor array under flow modulation. UCI Machine Learning Repository, 2015. URL <https://archive.ics.uci.edu/dataset/308/gas+sensor+array+under+flow+modulation>.
- Andrey Ziyatdinov, Jordi Fonollosa, Luis Fernández, Agustín Gutiérrez-Gálvez, Santiago Marco, and Alexandre Perera. Bioinspired early detection through gas flow modulation in chemo-sensory systems. *Sensors and Actuators B: Chemical*, 206:538–547, 2015a. doi: 10.1016/j.snb.2014.09.001.
- Andrey Ziyatdinov, Jordi Fonollosa, Luis Fernández, Agustín Gutiérrez-Gálvez, Santiago Marco, and Alexandre Perera. Data set from gas sensor array under flow modulation. *Data in Brief*, 3: 131–136, 2015b. doi: 10.1016/j.dib.2015.02.016.

A Reproduction and numerical verification

The MNIST experiments join the provided image archives, divide the pixels by 255, draw the test split before the training split, and use 128 training-only MiniBatchKMeans groups for the delta-membership factor. All 60,000 responses participate in each likelihood. The supplied fvGP coordinate-block helper assembles the final covariance; exact complete-component Cholesky solves determine the predictions. Group membership differs from the unspecified delta sets in the source paper. The initial feasibility search and rejected non-positive-definite training states are retained in the logs.

Table 2: Three-seed point-score reproduction of the MNIST digit-five task. Invalid variances are retained as computed; their distributional scores are not defined.

Seed	Brier	Invalid variances	Minimum variance	Time (s)	Peak RSS (GB)
11	0.012654	2	-0.00332	65.7	3.60
22	0.013138	16	-0.03133	67.5	3.61
33	0.012795	13	-0.04303	63.9	3.41

The mean raw squared score is 0.012862 ± 0.000249 , giving absolute difference 0.005138 from the published 0.018; it meets the stated coarse tolerance of 0.006. The score uses unbounded Gaussian-regression means against binary targets, without clipping the means to $[0, 1]$. This comparison must not be mistaken for a validated Gaussian predictive distribution. The polynomial taper evaluated with the Manhattan metric can yield invalid covariance behavior on these high-dimensional inputs; successful training factorizations alone do not guarantee valid conditional variances at new inputs. We report the raw scores and the failure counts, without clipping negative variances or assigning CRPS to them. This limitation pertains to this paper-style reproduction configuration, not to an evaluation of every possible delta mask. The chemical-sensing construction instead uses a one-dimensional temporal taper and is independently checked against the reference implementation.

On 200 training observations, the absolute likelihood difference from dense Cholesky is 1.51×10^{-11} , and the difference from the supplied fvGP gp2Scale calculation is 2.72×10^{-11} . Maximum predictive-mean and variance discrepancies are 5.28×10^{-14} and 1.23×10^{-15} , respectively. Vecchia with all predecessors agrees with the exact likelihood to 2.34×10^{-11} . The maximum derivative discrepancy divided by one plus the absolute analytic derivative is 8.27×10^{-7} . These independently computed checks are recorded in `logs/numerical_verification.json`.

B Hyperparameters and short-chain diagnostics

The exact models share a uniform prior over their recorded bounded parameterization. The periodic model uses three proposal blocks: $(\log v_d, \log r_d)$; $(\log v_o, \log r_o, \log P, \eta)$; and $\log \sigma$. The phase-removed model omits the inactive period from the second block. Wendland uses $(\log v, \log r)$ and $\log \sigma$. The final accepted state supplies all main-table predictions. The initial proposal standard deviation is recorded in each run configuration; it was reduced for the final adapted run following low acceptance in its first mode.

The bounds are $\log v_d \in [-6, 3]$, $\log v_o \in [-10, 3]$, $r_d \in [2, 120]$ seconds, $r_o \in [12, 120]$ seconds, $P \in [10, 14]$ seconds, $\eta \in [-3, 3]$, and $\log \sigma \in [-9, 0]$. Wendland uses the same drift-variance, drift-support, and noise bounds. The phase-removed model keeps the same bounds on its active parameters. These are restrictions on the statistical model, not physical measurements of sensor kinetics.

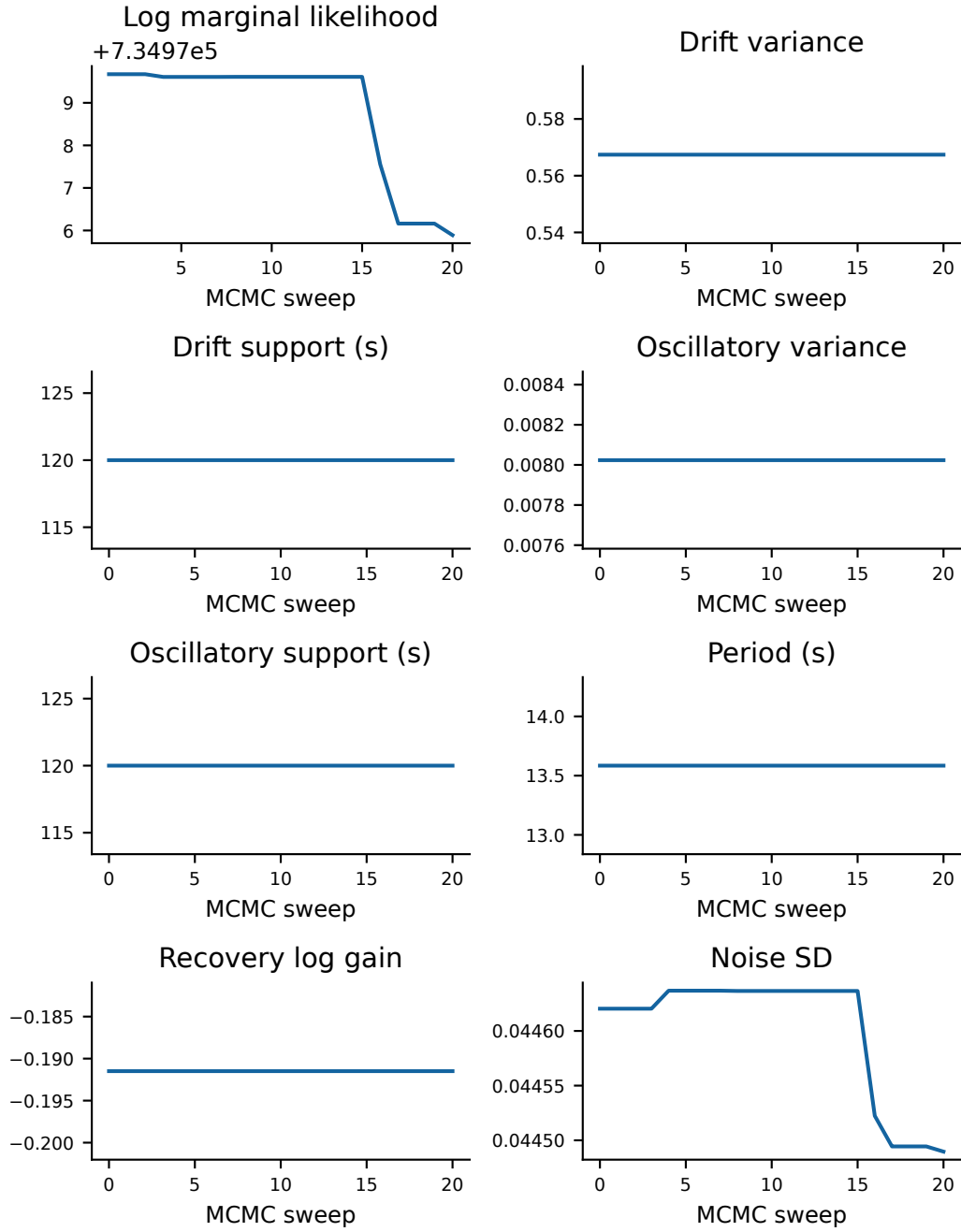


Figure 5: Final adapted-model likelihood and all hyperparameter trajectories. Repeated states are rejected proposals, not independent posterior samples. The drift support can remain at its prescribed upper bound.

Table 3: Fitted parameters. Variances and observation-noise standard deviations refer to training-standardized responses.

Parameter	Respiratory	Phase removed	Wendland
v_d	0.567408	0.018443	0.813460
r_d (s)	120.000000	13.801123	30.990229
v_o	0.008024	0.691515	absent
r_o (s)	120.000000	120.000000	absent
P (s)	13.584282	inactive	absent
η	-0.191492	-0.246008	absent
σ	0.044490	0.040863	0.040708

Table 4: Executed MCMC sweeps and acceptance by block, in the order described above. Low acceptance and short chains preclude a convergence claim.

Model	Sweeps	Block acceptance	Chain time (s)	Chain peak RSS (GB)
Respiratory	20	0.0%/0.0%/25.0%	395.4	93.28
Wendland	30	3.3%/6.7%	407.1	43.66
Phase removed	20	5.0%/0.0%/5.0%	497.4	116.01

C All seeds, regions, and initialization attempts

The records span five acquisition batches over four days. As a sensitivity to dependence between exposures in one batch, we enumerate all 3,125 resamplings of the five batches with replacement, retaining each batch in full. The adapted-minus-Wendland 95% percentile intervals are $[-0.00842, -0.00512]$ for RMSE and $[-0.00633, -0.00455]$ for CRPS. These are sensitivity checks with only five clusters, not precise estimates of uncertainty across new acquisition days. The source metadata and all batch scores are recorded in `logs/batch_bootstrap.json`.

Table 5: Every primary Vecchia seed uses the complete 500,000-point training pool and the same 9,280 held-out observations.

Run	RMSE	CRPS	Coverage (95%)	Wall time (h)
Vecchia 11	0.003516	0.002814	96.78%	0.153
Vecchia 22	0.003236	0.002701	96.77%	0.233
Vecchia 33	0.003804	0.003128	97.03%	0.136

The optimizer refinements were driven by training likelihood: an initially truncated coordinate search gave the richer kernel a worse objective than its nested simpler model, and the approximate fit subsequently identified a better exact-likelihood starting point. The additional starting-point diagnostic was performed after initial test scores were available; its selection rule was explicitly fixed to full exact training likelihood, and no test loss was passed to any optimizer. All completed scores, including the inferior initial fits, are retained below. The scientific kernel, frozen intervals, support bounds, and scoring rules were unchanged. These refinements are documented separately from the immutable original method plan.

With the seed-11 fitted covariance held fixed, the validation CRPS for 16, 32, 64, and 128 prediction parents is 0.00606, 0.00455, 0.00316, 0.00281, respectively. All four conditioning sizes also have logged full-data conditional likelihood evaluations. The primary three-seed baseline uses 128 parents, the strongest configuration in this validation diagnostic. Those scores are not retuned on the test set. The neighborhood diagnostic is saved in `logs/vecchia_refined_11/neighbor_screen.json`.

D Using every eligible training measurement

The all-eligible-data extension uses every archived measurement outside the frozen test and validation intervals. It keeps the original training-only normalization and kernel bounds, initializes from the

Table 6: Exposure, recovery, and gas-class breakdowns. R denotes the adapted respiratory kernel and W the existing Wendland kernel. Complete per-sensor and mixture-specific RMSE, CRPS, coverage curves, and baseline-seed breakdowns are included in `logs/subgroup_metrics.json`.

Region	Test n	RMSE (R)	RMSE (W)	CRPS (R)	CRPS (W)
Exposure	5,610	0.00352	0.01156	0.00260	0.00761
Recovery	3,670	0.00233	0.00729	0.00300	0.00855
Air	1,280	0.00027	0.00103	0.00022	0.00067
Acetone	2,400	0.00266	0.00705	0.00271	0.00743
Ethanol	2,400	0.00285	0.01213	0.00258	0.00795
Mixture	3,200	0.00406	0.01214	0.00394	0.01134

Table 7: Initialization sensitivity. Exact and Vecchia likelihood values are different objectives and should not be compared numerically across those two model classes.

Method	Initialization	RMSE	CRPS	Coverage (95%)	Final training objective
Wendland	Coordinate	0.00944	0.00797	96.48%	670061.5
Wendland	Gradient	0.01009	0.00798	96.44%	677218.2
Respiratory	First mode	0.00524	0.00452	96.81%	713387.5
Respiratory	Multiple starts	0.00310	0.00276	96.86%	734975.9
Vecchia (11)	Pilot	0.09335	0.03481	97.86%	371583.3
Vecchia (11)	Strong start	0.00352	0.00281	96.78%	730599.4

fitted exact model, and maximizes the full conditional-product likelihood separately for each seed. The training parent count remains 128; the prediction parent count is 512. At fixed initial covariance, validation CRPS for 128, 256, and 512 prediction parents is 0.00491, 0.00388, 0.00300, respectively. The lowest validation CRPS determines the prediction count before these fits start. A bounded full-likelihood noise initialization precedes joint coordinate optimization. The prespecified optimization cap is 30 minutes per seed, with at most 90 joint objective calls in addition to noise initialization. Ten single-threaded workers serve each run; three runs are concurrent. A rolling Cholesky deletion and positive rank-one update computes the same parent conditionals as fresh factorizations. Its numerical comparison is recorded in `logs/rolling_vecchia_verification.json`. This extension was specified after the initial paper audit and is not part of the original matched-pool preregistration. It adds data but does not change any held-out interval or response.

Executed fits: seed 11: 98 full likelihood evaluations, Maximum number of function evaluations has been exceeded; seed 22: 98 full likelihood evaluations, Maximum number of function evaluations has been exceeded; seed 33: 98 full likelihood evaluations, Maximum number of function evaluations has been exceeded. Optimizer budgets and these starts do not establish convergence.

Joint fitting substantially worsened validation CRPS relative to the initializer. A separately recorded validation-only rule therefore compared the best full-data noise-initialization state with the final joint state. All three seeds selected the earlier state. Thus the selected covariance parameters are transferred from the exact-model checkpoint with the recorded seed perturbation, while noise is fitted by the complete 6,403,200-point Vecchia likelihood. The main comparison uses these validation-selected states. The full joint fits remain reported as ablations below, including their unfavorable CRPS and complete coverage curves. Selection never uses their test losses. The rule is saved in `logs/all_eligible_checkpoint_rule.json`. Selected-run times include the complete joint fitting/evaluation cost plus the extra checkpoint validation and test evaluation; they additionally inherit the shared exact initialization cost.

Among conditionals with 128 parents, the median predecessor time span changes from 71.00 seconds in the matched pool to 5.12 seconds in the full archive. A fixed parent count is therefore not a fixed physical context. The joint-fit oscillatory variance shrinks to its lower bound or very near it, while drift variance approaches its upper bound. The parameter snapshots and

logs/parent_span_summary.json document this change; they do not establish that every larger-neighborhood fit would fail.

Table 8: Vecchia using all 6,403,200 eligible observations, evaluated on the unchanged 9,280 test targets. Both validation-selected and final joint-fitted states are retained for every seed.

Checkpoint	Seed	RMSE	CRPS	Coverage (95%)	Time (h)	Peak RSS (GB)
Selected	11	0.004090	0.002875	94.19%	0.537	0.76
Selected	22	0.004440	0.002955	93.79%	0.537	0.76
Selected	33	0.003888	0.002828	94.47%	0.537	0.74
Joint fitted	11	0.011029	0.036934	98.47%	0.473	0.76
Joint fitted	22	0.010617	0.035942	98.45%	0.474	0.76
Joint fitted	33	0.011242	0.037501	98.47%	0.474	0.74

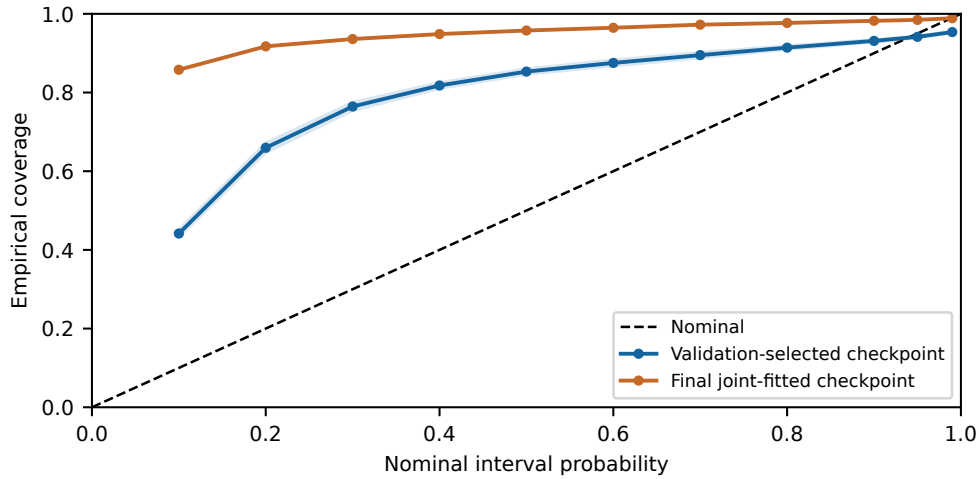


Figure 6: Coverage curves for both full-archive Vecchia checkpoints. Bands show standard deviations across the three seeds. Validation selection improves predictive scores but does not eliminate miscalibration.

E Support perturbations and resource accounting

To separate support from parameter refitting, we perturb only the fitted drift support while holding every other parameter fixed. The primary model retains its preregistered support bound. The larger-radius diagnostic is a post-fit sensitivity check, not a selected model. Every configuration recomputes the full 500,000-point covariance, solve, and log determinant and predicts the same held-out targets.

Table 9: Fixed-parameter support perturbations. Coverage denotes empirical 95% interval coverage; runtime and sampled peak RSS include prediction.

r_d (s)	Nonzeros (%)	RMSE	CRPS	Coverage	Time (s)	Peak RSS (GB)
30	0.06836	0.00970	0.00831	97.18%	25.6	7.38
60	0.06836	0.00396	0.00424	96.65%	22.6	9.54
120	0.06836	0.00310	0.00276	96.86%	21.9	11.79
240	0.10309	0.00440	0.00268	96.12%	23.1	17.55

The primary adapted, existing-kernel, and phase-removed runs use 16, 4, and 8 Dask workers, respectively, with one numerical-library thread each. The three primary Vecchia runs use 8, 4, and 8 workers. Workers are threads in a shared-memory Dask cluster. Separate jobs were sometimes

concurrent, so timings are descriptive, not controlled throughput benchmarks. Peak memory is sampled process RSS, including children, at 0.1-second intervals. The final adapted run takes 827.4 seconds and peaks at 95.37 GB. Its test prediction takes 15.8 seconds with sampled process RSS 94.78 GB. The main table excludes reused initialization stages: the earlier completed full-data exact fit took 1281.7 seconds (peak 162.54 GB), and the seed-11 Vecchia refinement supplying the final exact start took 552.2 seconds. Consequently neither the final-run timings nor Vecchia’s small refinement RSS represent independent cold-start costs. The complete trace includes these stages and the unsuccessful starts. The final exact solve has relative residual 2.17×10^{-14} ; its standardized-data log marginal likelihood is 734975.890438, with log determinant -2891538.680303. Full hyperparameter optimization, likelihood checks, MCMC proposals within the prior bounds, and final prediction use the complete training set.

F Training-only spectral evidence

For each trajectory and phase, we project the training observations onto a cubic time trend and sinusoidal terms over a fixed grid of candidate periods. The reported quantity is the fraction of residual variance explained after removing the cubic trend. Exposure uses the interior interval 24–168 seconds; recovery uses 192–276 seconds, avoiding the source-removal transition. The frequency diagnostic uses no test responses and does not tune the reported GP. Experiment-cluster bootstrap intervals, trajectory spectra, and the exact period grid are supplied with the artifact.

Table 10: Peak training-only harmonic variance fractions after cubic detrending, averaged across trajectories within each gas class. These are variance fractions, not chemical detection rates.

Gas	Exposure period (s)	Exposure fraction	Recovery period (s)	Recovery fraction
Air	13.55	0.582	13.55	0.586
Acetone	13.85	0.153	13.65	0.708
Ethanol	13.55	0.269	13.70	0.716
Mixture	13.55	0.240	13.70	0.716

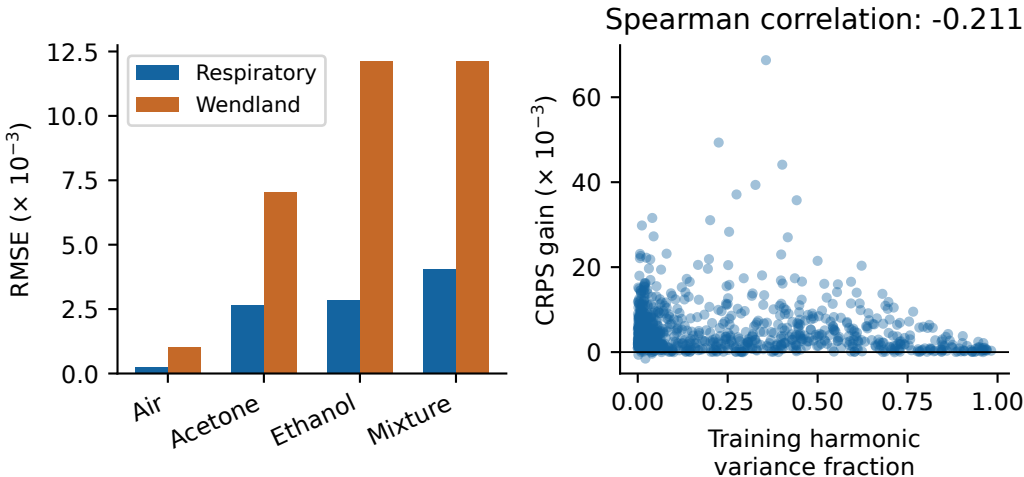


Figure 7: Left: errors by broad gas class. Right: trajectory-level CRPS improvement over Wendland against harmonic structure measured using training observations only. Positive improvement favors the adapted covariance.

G Artifact and provenance

The archive is distributed under the UCI dataset’s CC BY 4.0 license. The data retrieval URL, archive and extracted-file hashes, preprocessing, all split identities, and their measured sizes are saved in

`logs/data_retrieval.json` and `logs/data_preparation.json`. The original method plan was committed before any full-data fit; optimizer and backend refinements are separately recorded. Source snapshots and their licenses are retained alongside the supplied-library checksums.

The distinct kernel is implemented in `code/respiratory_kernel.py`. Sparse assembly and exact component factorization are in `code/sparse_engine.py`; exact gradients are in `code/likelihood_gradient.py`. Entry points cover retrieval, preparation, MNIST reproduction, exact fitting, Vecchia fitting, numerical checks, support diagnostics, spectral analysis, and figure/table generation. The final tables are generated from saved predictions and numerical logs, and are cross-checked by recomputing their scores. `README.md` documents commands and the complete execution history; `attempts_log.md` documents unsuccessful starts and dependency failures. No dataset other than the public chemical-sensor archive was acquired for the new-domain experiment.